Experiment Report of "Multimodal sheaves on graphs"

This solution addresses the embedding and aggregation needs of multimodal content recommendation within the VK ecosystem. Combining Cellular Sheaf theory and graph neural network technology, it designs a complete implementation process from unimodal graph sheaf learning to multimodal fusion. It adapts to multimodal (video, text, title, audio) content embedding and aggregation scenarios in recommendation systems, and ultimately delivers a working model prototype, experimental reports, and related metrics/hyperparameter tables to meet the technical requirements of multimodal recommendation within the VK ecosystem.

I. Project Goals and Objectives

Core Objective

Develop a working prototype of a graph learning model based on cellular sheaf theory to achieve effective aggregation of multimodal embeddings on graphs. It can integrate content embedding features from different modalities such as video, text, title, and voice, providing representative multimodal joint embeddings for recommendation systems in the VK ecosystem, thereby improving the relevance and accuracy of recommended content.

Specific goals

1. Master the core practical methods of graph beam theory, implement the graph beam flow learning algorithm for single-modal data, and complete feature message passing and aggregation;
2. Extend the unimodal graph bundle model to multimodal scenarios, design cross-modal graph bundle mapping rules, and enable the embeddings of different modalities to interact and merge through common bundle mapping;
3. The model was trained, hyperparameters were tuned, and performance was validated on an open recommendation dataset, verifying the feasibility and superiority of the graph bundle method in multimodal embedding aggregation.

2. Core Theories and Methods

Based on the theory of cellular sheaf, and combined with the message passing mechanism of graph neural networks (GNNs), the model was designed with reference to the core ideas of "Neural Sheaf Diffusion":

- Single-modal graph bundle learning: Constructs a graph bundle structure for embedding of a single modality, assigns a vector space (stalk space) to each node (content)/edge (content association) of the graph, defines linear restriction maps of nodes and edges, and realizes feature diffusion and aggregation through the Sheaf Laplacian operator;
- Multimodal graph bundle fusion: Independent stem spaces are designed for different modalities + shared cross-modal bundle mapping. The embeddings of video, text, title, and speech are mapped to their respective node stem spaces. Cross-modal feature interaction is achieved through edge bundle mapping, and global aggregation of multimodal embeddings is completed through the bundle diffusion process.
- Model training strategy: An end-to-end training approach is adopted, with the RMSE of the recommendation task as the optimization objective, while simultaneously learning the bundle mapping parameters and cross-modal fusion parameters of each modality.

Special note: The final code achieves single/multimodal compatibility. When the modal parameter is set to single-mode, single-mode experiments can be performed.

3. Dataset

3.1 MovieLens-1M

- Rating history: 1 million records
- Number of users: 6040
- Number of movies: 3706
- Rating range: 1~5
- Division: 8:1:1 (Training/Validation/Testing)

3.2 Multimodal features

- Text features: Movie synopsis (BERT 768-dimensional)
- Video/Image features: Poster (ResNet 2048-dimensional)
- Features have been pre-extracted and aligned

4. Model Structure (MMSRec)

4.1 Embedding Layer

User Embedding, Item Embedding, Modal Feature Projection

4.2 Graph Stalk Layer (Core)

Node Stalk Space

Edge Stalk Space

Graph Diffusion Layer: Achieving More Robust Higher-Order Information Propagation

4.3 Multimodal Fusion Layer

Supports Three Fusion Strategies:

Concat
Average
Attention

4.4 Prediction Layer

Feature Fusion → Fully Connected Layer → Rating Prediction

5. Experimental setup

5.1 Environment

Python 3.10

PyTorch 2.x

CUDA 11.7+

GPU: NVIDIA GPU RTX 4060 Laptop

5.2 Basic Parameters

Optimizer: Adam

Random Seed: 42

Early Stop: Patience=3

Maximum Number of Rounds: 30 (tuning)、50 (training)

Evaluation Metrics: RMSE

6. Automated hierarchical hyperparameter tuning

6.1 Tuning strategy

Employing a three-stage fully automated hierarchical search:

Core Stage
Learning rate, node dimension, fusion method, batch size

Secondary Stage
Edge dimension, number of graph diffusion layers, regularization, dropout
Fine Stage
Local fine-tuning search

6.2 Search space

Core Stage

- lr: 1e-4, 3e-4, 5e-4, 8e-4, 1e-3
- node_stalk_dim: 32, 64, 128, 256
- fusion_type: attention / concat / average
- batch_size: 64, 128, 256

Secondary Stage

- edge_stalk_dim: 32, 64, 128
- num_diffusion_layers: 1,2,3,4
- weight_decay: 1e-5,1e-4,1e-3
- dropout: 0.0,0.1,0.2,0.3

Fine Stage

- lr: best ±20%
- node_stalk_dim: best ±64

6.3 Engineering Optimization (Ensuring Speed and Stability)

Using 20% data sampling during optimization improves speed by 5-8 times.

Mixed precision training.

Early stopping mechanism.

Automatic reading of optimal configuration and automatic parameter transfer.

Automatic saving of logs, curves, and results.

7. Results

7.1 Final optimal model configuration

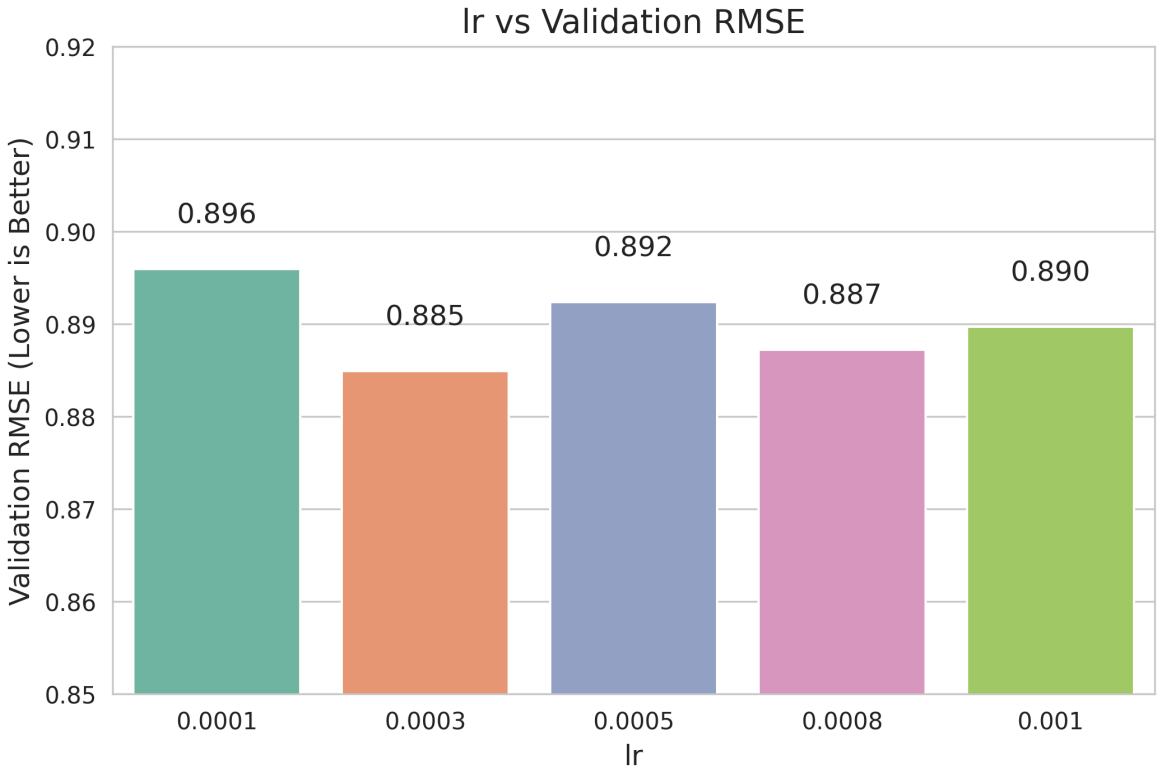
config

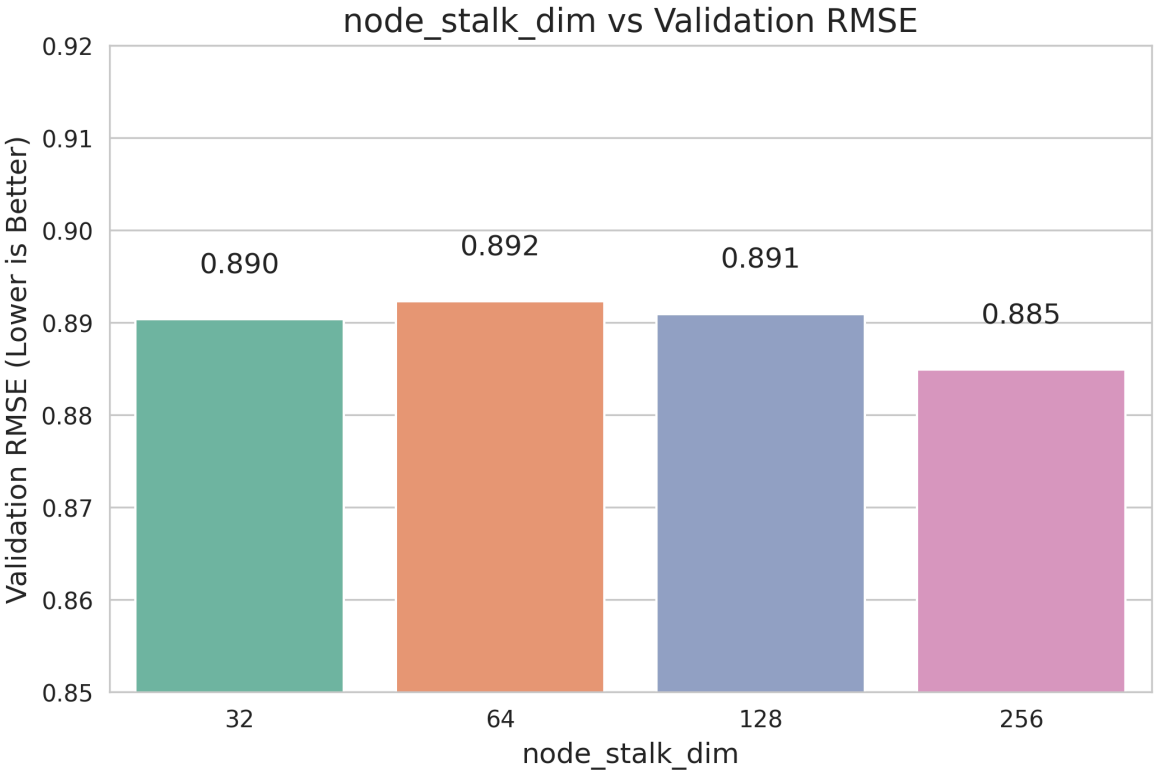
lr: 0.0003
node_stalk_dim: 256
fusion_type: concat
batch_size: 256
edge_stalk_dim: 32
num_diffusion_layers: 1
weight_decay: 1e-05
dropout_rate: 0.1

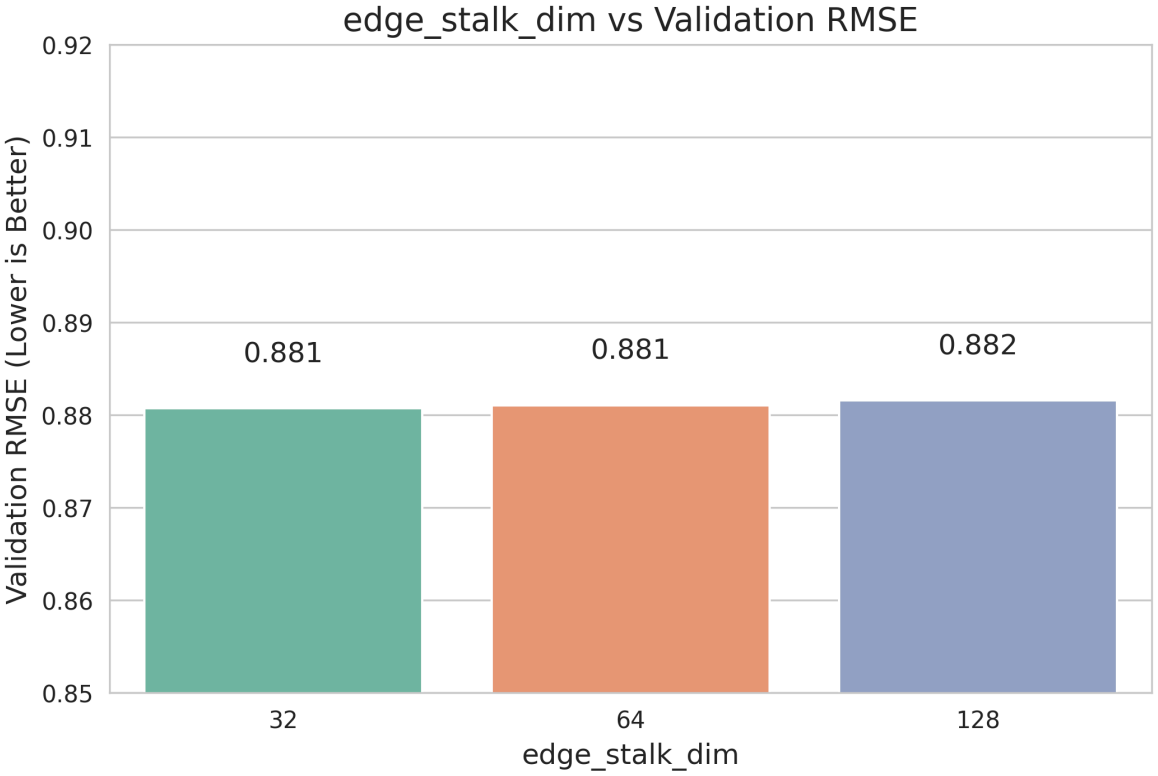
metrics

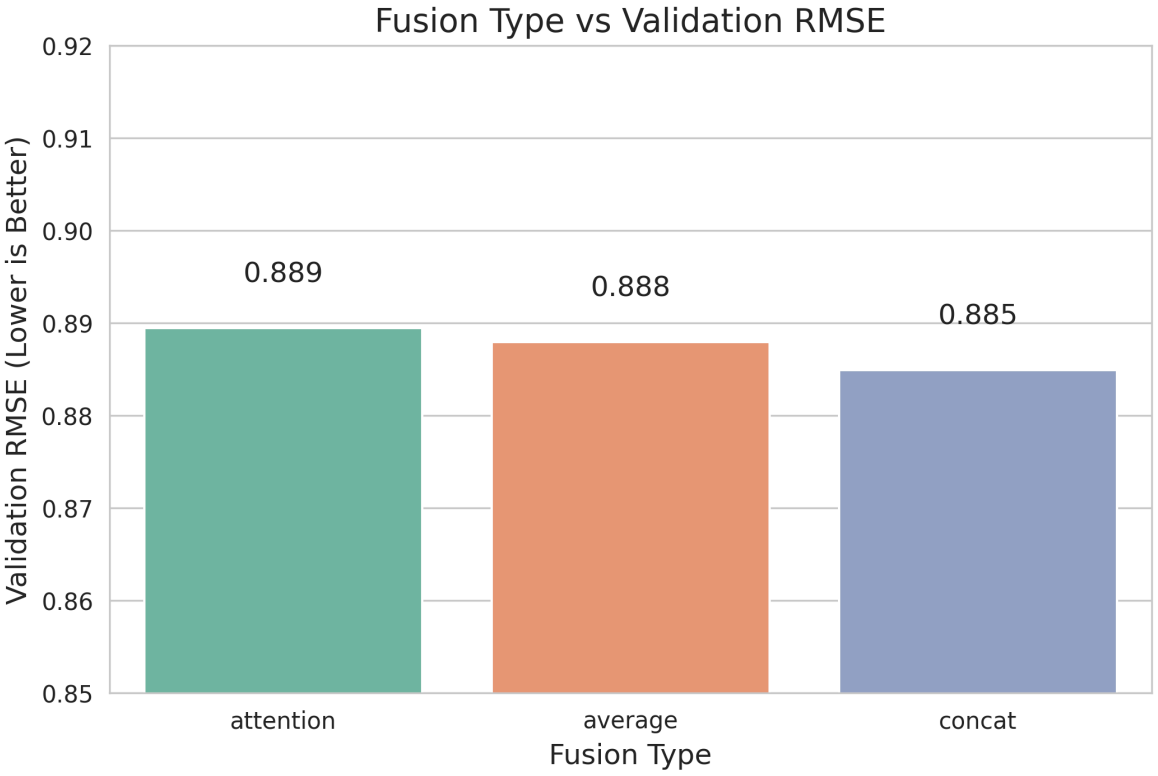
RMSE：0.8560

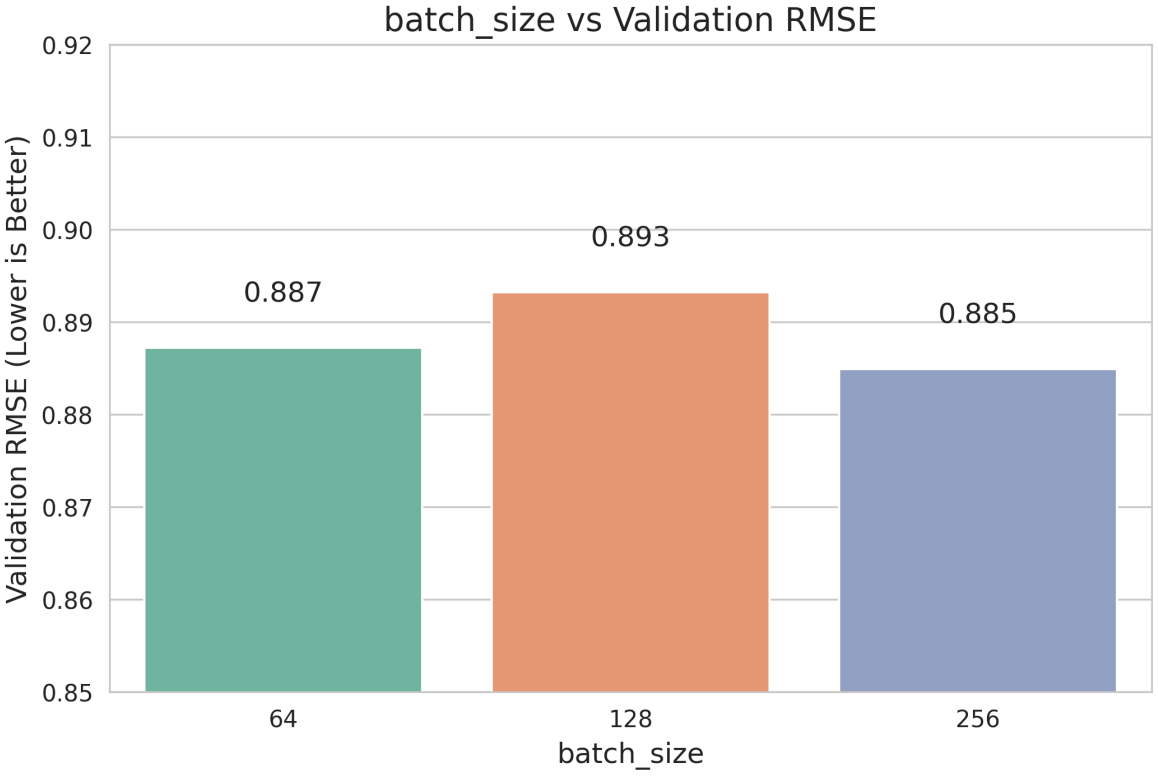
7.2 Visualization results

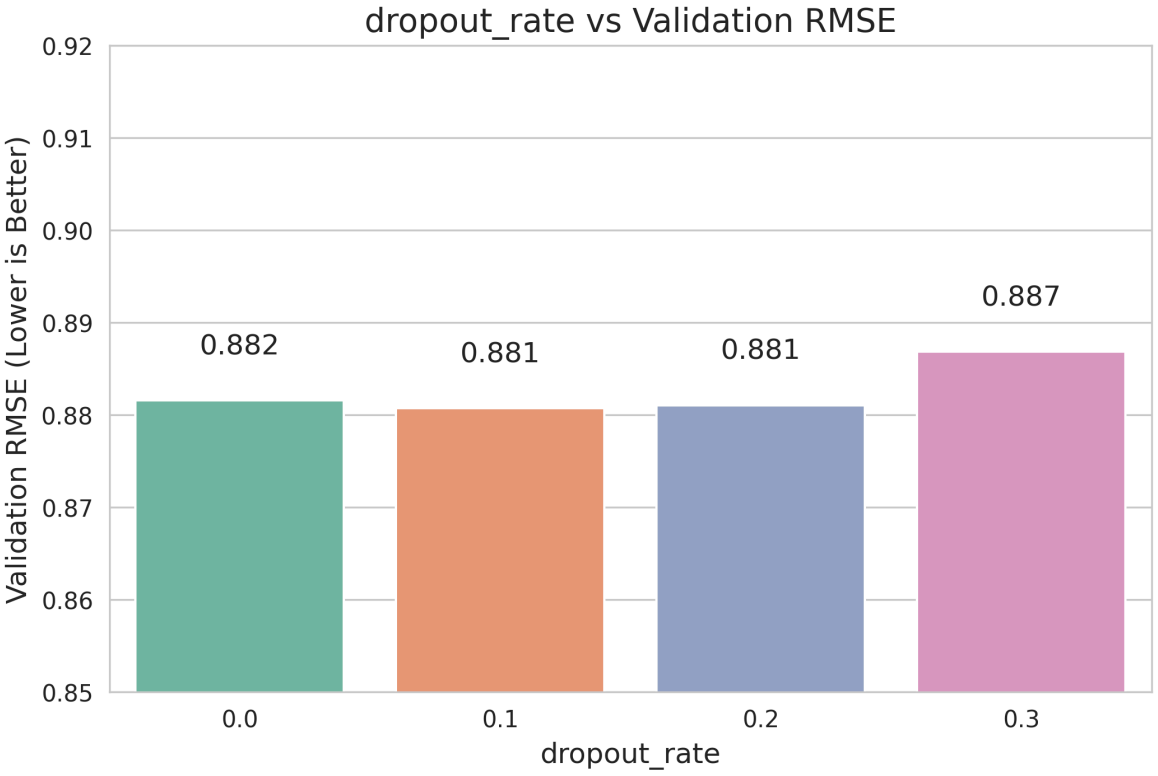


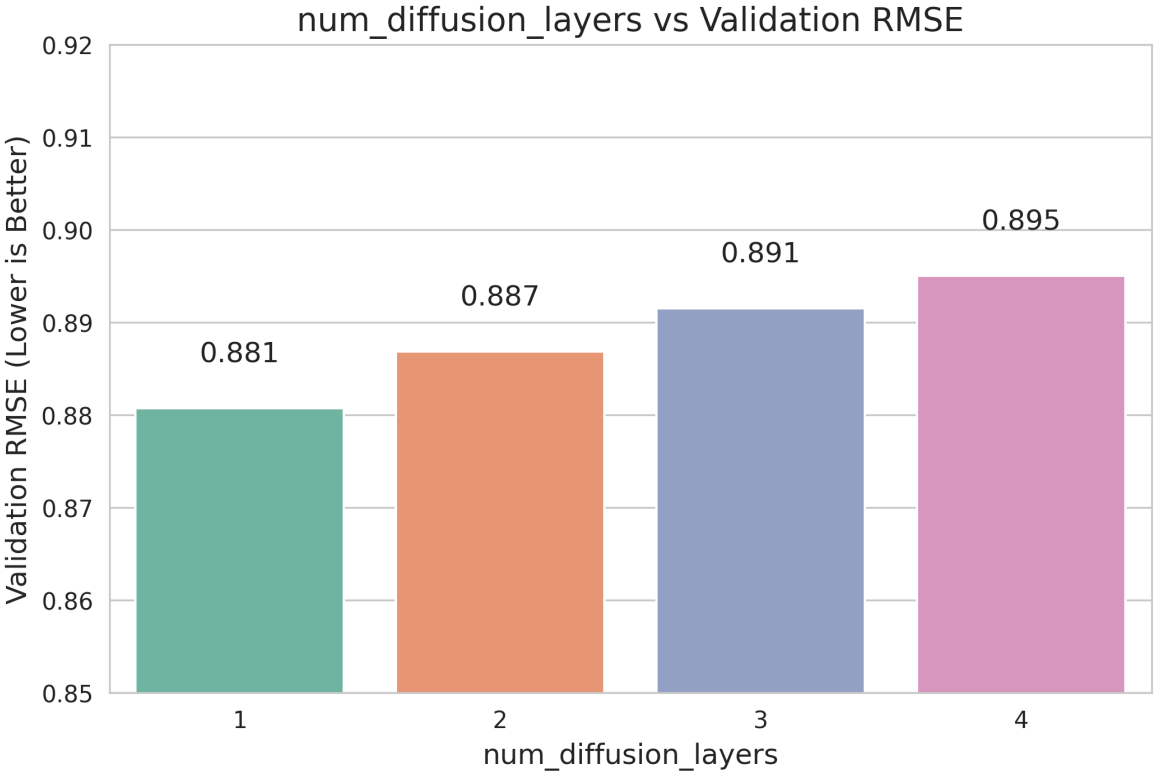

node_stalk_dim vs Validation RMSE
Validation RMSE (Lower is Better)
0.92
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.890
0.892
0.891
0.885
32
64
128
256
node_stalk_dim


edge_stalk_dim vs Validation RMSE
Validation RMSE (Lower is Better)
0.92
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.881
0.881
0.882
32
64
128
edge_stalk_dim


Fusion Type vs Validation RMSE
Validation RMSE (Lower is Better)
0.92
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.889
0.888
0.885
attention
average
concat
Fusion Type


batch_size vs Validation RMSE
Validation RMSE (Lower is Better)
0.92
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.887
0.893
0.885
64
128
256
batch_size


dropout_rate vs Validation RMSE
Validation RMSE (Lower is Better)
0.92
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.882
0.881
0.881
0.887
0.0
0.1
0.2
0.3
dropout_rate


num_diffusion_layers vs Validation RMSE
Validation RMSE (Lower is Better)
0.92
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.881
0.887
0.891
0.895
1
2
3
4
num_diffusion_layers

weight_decay vs Validation RMSE
Validation RMSE (Lower is Better)
0.92
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.881
0.882
0.895
1e-05
0.0001
0.001
weight_decay